

COLUMN REBAR ANALYSIS REPORT

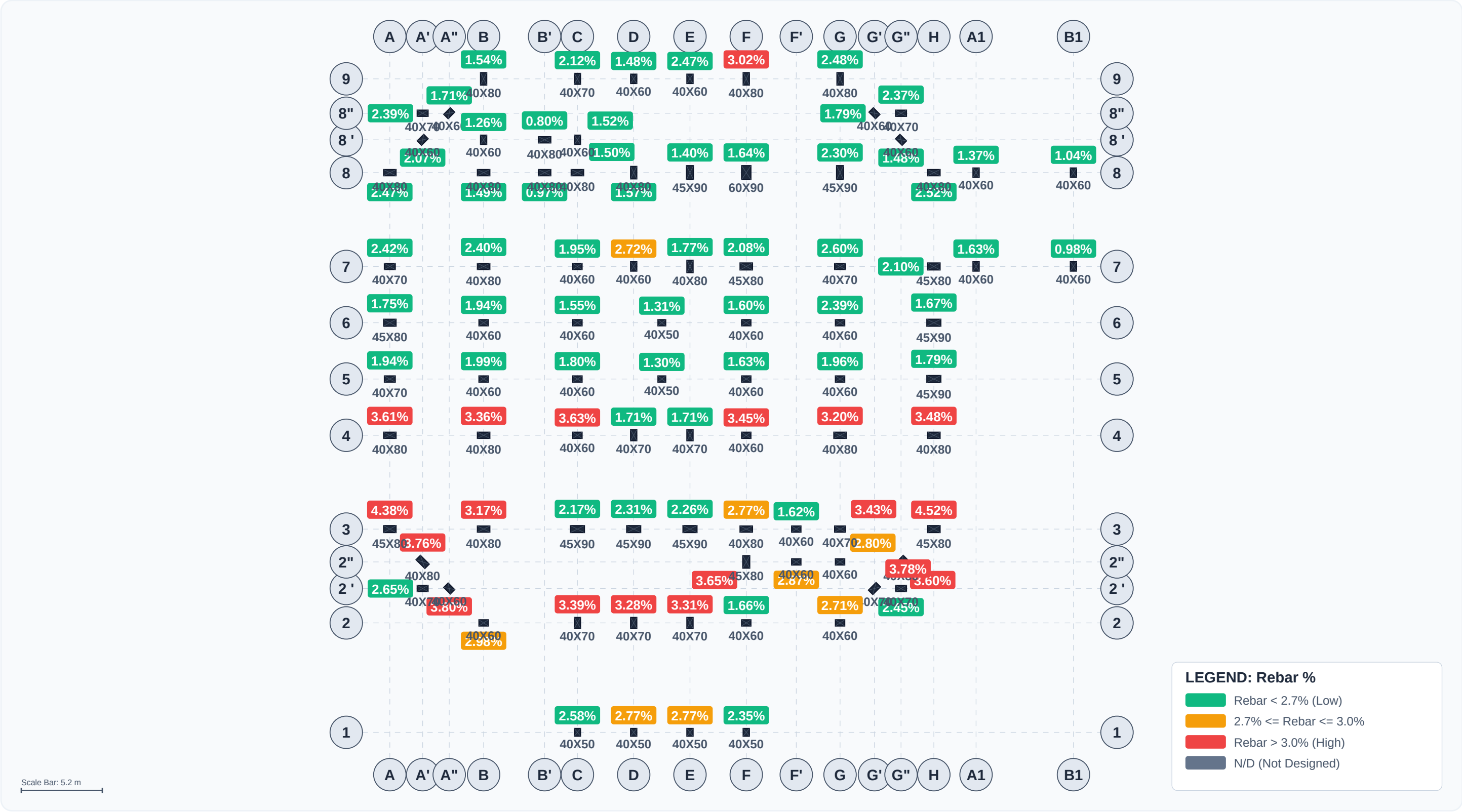
Programmatic Extraction & Visualization of Longitudinal Reinforcement from ETABS Models

Project Name:	Reinforcement Column Audit
Analysis Software:	CSI ETABS (Active API Interface)
Date Generated:	May 24, 2026
Status:	COMPLETED

Executive Summary: This report presents the design results of columns extracted directly from the active ETABS structural analysis model. It maps individual column frame elements across various stories to their closest horizontal grid points using ETABS bay and connectivity definition tables, and computes the **maximum longitudinal rebar percentage** required at each grid location from the design PMM Envelope. The layout drawings on the subsequent page display the physical shape (rectangular/circular), size, and orientation (local coordinate angle) of the columns. The colors indicate critical reinforcement demands, helping engineers instantly inspect and audit reinforcement concentrations in the model.

Column Key Plan & Rebar Distribution

The drawing below shows column geometries in plan (exaggerated scale for visibility) rotated based on local axes, alongside color-coded badges indicating maximum rebar percentage. Standard AABB collision-avoidance logic is implemented to eliminate overlap of label tags.



Detailed Reinforcement Data Table

Summary of all column coordinates, section details, orientations, and the maximum rebar percentage with governing structural locations.

Grid Point	X Coord (m)	Y Coord (m)	Drawn Section	Shape	Dimensions (mm)	Angle (°)	Max Rebar %	Governing Story	Frame Element
A"-2 '	3.800	2.200	C40X60	Rectangular	400 x 600	135.0	3.80%	GF	1489
A"-8"	3.800	32.600	C40X60	Rectangular	400 x 600	45.0	1.71%	2F	1747
A'-2 '	2.100	2.200	C40X70	Rectangular	400 x 700	0.0	2.65%	3F	1496
A'-2"	2.100	3.900	C40X80	Rectangular	400 x 800	135.0	3.76%	GF	1494
A'-8 '	2.100	30.900	C40X60	Rectangular	400 x 600	45.0	2.07%	1F	1508
A'-8"	2.100	32.600	C40X70	Rectangular	400 x 700	0.0	2.39%	ROOF	1514
A-3	0.000	6.000	C45X80	Rectangular	450 x 800	540.0	4.38%	GF	1075
A-4	0.000	12.000	C40X80	Rectangular	400 x 800	0.0	3.61%	1F	557
A-5	0.000	15.600	C40X70	Rectangular	400 x 700	0.0	1.94%	1F	566
A-6	0.000	19.200	C45X80	Rectangular	450 x 800	180.0	1.75%	1F	565
A-7	0.000	22.800	C40X70	Rectangular	400 x 700	0.0	2.42%	ROOF	689
A-8	0.000	28.800	C40X80	Rectangular	400 x 800	0.0	2.47%	1F	563
A1-7	37.500	22.800	C40X60	Rectangular	400 x 600	90.0	1.63%	1B	1762
A1-8	37.500	28.800	C40X60	Rectangular	400 x 600	90.0	1.37%	1B	1708
B'-8	9.900	28.800	C40X80	Rectangular	400 x 800	0.0	0.97%	TOWER	804
B'-8 '	9.900	30.900	C40X80	Rectangular	400 x 800	0.0	0.80%	TOWER	1109
B-2	6.000	0.000	C40X60	Rectangular	400 x 600	180.0	2.98%	1F	541
B-3	6.000	6.000	C40X80	Rectangular	400 x 800	180.0	3.17%	GF	1073
B-4	6.000	12.000	C40X80	Rectangular	400 x 800	0.0	3.36%	1F	1174
B-5	6.000	15.600	C40X60	Rectangular	400 x 600	0.0	1.99%	3F	658
B-6	6.000	19.200	C40X60	Rectangular	400 x 600	0.0	1.94%	3F	655
B-7	6.000	22.800	C40X80	Rectangular	400 x 800	0.0	2.40%	3F	654
B-8	6.000	28.800	C40X80	Rectangular	400 x 800	0.0	1.49%	2F	1721
B-8 '	6.000	30.900	C40X60	Rectangular	400 x 600	90.0	1.26%	1F	1633
B-9	6.000	34.800	C40X80	Rectangular	400 x 800	90.0	1.54%	1F	1630

Grid Point	X Coord (m)	Y Coord (m)	Drawn Section	Shape	Dimensions (mm)	Angle (°)	Max Rebar %	Governing Story	Frame Element
B1-7	43.730	22.800	C40X60	Rectangular	400 x 600	90.0	0.98%	1B	1764
B1-8	43.730	28.800	C40X60	Rectangular	400 x 600	90.0	1.04%	1B	1756
C-1	12.000	-7.000	C40X50	Rectangular	400 x 500	90.0	2.58%	GF	939
C-2	12.000	0.000	C40X70	Rectangular	400 x 700	270.0	3.39%	GF	1079
C-3	12.000	6.000	C45X90	Rectangular	450 x 900	180.0	2.17%	GF	1074
C-4	12.000	12.000	C40X60	Rectangular	400 x 600	0.0	3.63%	1F	558
C-5	12.000	15.600	C40X60	Rectangular	400 x 600	0.0	1.80%	1F	559
C-6	12.000	19.200	C40X60	Rectangular	400 x 600	0.0	1.55%	1F	560
C-7	12.000	22.800	C40X60	Rectangular	400 x 600	0.0	1.95%	1F	561
C-8	12.000	28.800	C40X80	Rectangular	400 x 800	0.0	1.50%	1F	562
C-8 '	12.000	30.900	C40X60	Rectangular	400 x 600	90.0	1.52%	1F	998
C-9	12.000	34.800	C40X70	Rectangular	400 x 700	90.0	2.12%	1F	551
D-1	15.600	-7.000	C40X50	Rectangular	400 x 500	90.0	2.77%	1F	538
D-2	15.600	0.000	C40X70	Rectangular	400 x 700	270.0	3.28%	GF	1080
D-3	15.600	6.000	C45X90	Rectangular	450 x 900	180.0	2.31%	GF	1070
D-4	15.600	12.000	C40X70	Rectangular	400 x 700	270.0	1.71%	GF	1811
D-7	15.600	22.800	C40X60	Rectangular	400 x 600	90.0	2.72%	ROOF	710
D-8	15.600	28.800	C40X80	Rectangular	400 x 800	90.0	1.57%	1F	553
D-9	15.600	34.800	C40X60	Rectangular	400 x 600	90.0	1.48%	1F	552
E-1	19.200	-7.000	C40X50	Rectangular	400 x 500	90.0	2.77%	1F	539
E-2	19.200	0.000	C40X70	Rectangular	400 x 700	270.0	3.31%	GF	1081
E-3	19.200	6.000	C45X90	Rectangular	450 x 900	180.0	2.26%	GF	1071
E-4	19.200	12.000	C40X70	Rectangular	400 x 700	270.0	1.71%	GF	1813
E-7	19.200	22.800	C40X80	Rectangular	400 x 800	90.0	1.77%	1F	555
E-8	19.200	28.800	C45X90	Rectangular	450 x 900	90.0	1.40%	1F	967
E-9	19.200	34.800	C40X60	Rectangular	400 x 600	90.0	2.47%	1F	953
F'-2"	26.000	3.900	C40X60	Rectangular	400 x 600	0.0	2.87%	MF	1612
F'-3	26.000	6.000	C40X60	Rectangular	400 x 600	0.0	1.62%	GF	1077
F-1	22.800	-7.000	C40X50	Rectangular	400 x 500	90.0	2.35%	GF	942

Grid Point	X Coord (m)	Y Coord (m)	Drawn Section	Shape	Dimensions (mm)	Angle (°)	Max Rebar %	Governing Story	Frame Element
F-2	22.800	0.000	C40X60	Rectangular	400 x 600	180.0	1.66%	GF	1082
F-2"	22.800	3.900	C45X80	Rectangular	450 x 800	270.0	3.65%	GF	1170
F-3	22.800	6.000	C40X80	Rectangular	400 x 800	0.0	2.77%	1F	584
F-4	22.800	12.000	C40X60	Rectangular	400 x 600	0.0	3.45%	1F	583
F-5	22.800	15.600	C40X60	Rectangular	400 x 600	0.0	1.63%	1F	582
F-6	22.800	19.200	C40X60	Rectangular	400 x 600	0.0	1.60%	1F	581
F-7	22.800	22.800	C45X80	Rectangular	450 x 800	0.0	2.08%	1F	580
F-8	22.800	28.800	C60X90	Rectangular	600 x 900	90.0	1.64%	1B	976
F-9	22.800	34.800	C40X80	Rectangular	400 x 800	90.0	3.02%	1F	960
G"-2 '	32.700	2.200	C40X70	Rectangular	400 x 700	0.0	2.45%	ROOF	805
G"-2"	32.700	3.900	C40X80	Rectangular	400 x 800	45.0	3.60%	GF	1022
G"-8 '	32.700	30.900	C40X60	Rectangular	400 x 600	135.0	1.48%	2F	1115
G"-8"	32.700	32.600	C40X70	Rectangular	400 x 700	0.0	2.37%	ROOF	1127
G'-2 '	31.000	2.200	C40X70	Rectangular	400 x 700	45.0	3.78%	GF	1151
G'-8"	31.000	32.600	C40X60	Rectangular	400 x 600	135.0	1.79%	1F	1123
G-2	28.800	0.000	C40X60	Rectangular	400 x 600	180.0	2.71%	GF	1107
G-2"	28.800	3.900	C40X60	Rectangular	400 x 600	0.0	2.80%	GF	1169
G-3	28.800	6.000	C40X70	Rectangular	400 x 700	180.0	3.43%	GF	1072
G-4	28.800	12.000	C40X80	Rectangular	400 x 800	0.0	3.20%	1F	576
G-5	28.800	15.600	C40X60	Rectangular	400 x 600	0.0	1.96%	3F	669
G-6	28.800	19.200	C40X60	Rectangular	400 x 600	0.0	2.39%	3F	670
G-7	28.800	22.800	C40X70	Rectangular	400 x 700	0.0	2.60%	3F	673
G-8	28.800	28.800	C45X90	Rectangular	450 x 900	90.0	2.30%	1F	988
G-9	28.800	34.800	C40X80	Rectangular	400 x 800	90.0	2.48%	ROOF	978
H-3	34.800	6.000	C45X80	Rectangular	450 x 800	540.0	4.52%	GF	1076
H-4	34.800	12.000	C40X80	Rectangular	400 x 800	0.0	3.48%	1F	574
H-5	34.800	15.600	C45X90	Rectangular	450 x 900	0.0	1.79%	1F	573
H-6	34.800	19.200	C45X90	Rectangular	450 x 900	180.0	1.67%	1F	572
H-7	34.800	22.800	C45X80	Rectangular	450 x 800	0.0	2.10%	1F	571

Grid Point	X Coord (m)	Y Coord (m)	Drawn Section	Shape	Dimensions (mm)	Angle (°)	Max Rebar %	Governing Story	Frame Element
H-8	34.800	28.800	C40X80	Rectangular	400 x 800	0.0	2.52%	1F	570
Off-Grid-5	17.400	15.600	C40X50	Rectangular	400 x 500	0.0	1.30%	GF	1465
Off-Grid-6	17.400	19.200	C40X50	Rectangular	400 x 500	0.0	1.31%	GF	1466